

# KATHRYN B. NEWHART

kathryn.newhart@oregonstate.edu

## PROFESSIONAL EXPERIENCE

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| <b>Oregon State University</b><br><i>Assistant Professor of Environmental Engineering</i>             | June 2024 - Present<br><i>Corvallis, OR</i>    |
| <b>United States Military Academy</b><br><i>Assistant Professor of Environmental Engineering</i>      | June 2021 - June 2024<br><i>West Point, NY</i> |
| <b>Metro Wastewater Reclamation District</b><br><i>Technology &amp; Innovation Engineer Associate</i> | March 2020 - May 2021<br><i>Denver, CO</i>     |

## EDUCATION

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| <b>Doctor of Philosophy</b><br><i>Civil and Environmental Engineering</i>                                                                                                                                        | 2018 - 2020<br><i>Colorado School of Mines, Golden, CO</i> |
| <ul style="list-style-type: none"><li>· Dissertation: "Data-driven process control of municipal wastewater treatment"</li><li>· Advisors: Prof. Tzahi Cath and Prof. Amanda Hering (Baylor University)</li></ul> |                                                            |
| <b>Master of Science</b><br><i>Civil and Environmental Engineering</i>                                                                                                                                           | 2016 - 2018<br><i>Colorado School of Mines, Golden, CO</i> |
| <b>Bachelor of Science</b><br><i>Environmental Engineering</i>                                                                                                                                                   | 2013 - 2016<br><i>Colorado School of Mines, Golden, CO</i> |

## GRANTS AND AWARDS

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"Integrated Computer Vision Intelligence System for Activated Sludge Microscopy and Process Optimization," Water Research Foundation, Tailored Collaboration Program, 5389, Awarded 2026. Total award \$300,000. OSU award \$175,000. Principal Investigator.

"Digital Water for DOD Installations: Demonstration and Deployment Guidance for Leveraging Big Data at Water and Wastewater Utilities," U.S. Department of Defense, Environmental Security Technology Certification Program, Awarded 2025. Total award \$1,010,064. OSU award \$589,914. Principal Investigator.

"Increasing Installation Energy Resilience through Energy Producing Resource Recovery Facilities," U.S. Department of Defense, Environmental Security Technology Certification Program, Awarded 2025. Total award \$1,566,825. OSU award \$25,050. co-Principal Investigator.

"Unlocking the Nationwide Potential of Water Reuse," U.S. Environmental Protection Agency, National Priorities: Water Innovation, Science, and Engagement to Advance Water Reuse. EPA-G2021-ORD&EI. Awarded 2023. Total award \$4,000,000. Research Partner at West Point.

"Data-driven Fault Detection and Process Control for Potable Reuse with Reverse Osmosis," National Alliance for Water Innovation, Autonomous Water and Precision Separations. NAWI-2-2021. Awarded 2022. Total award \$1,034,000. co-Principal Investigator.

"Crossing the Finish Line: Integration of Data-Driven Process Control for Maximization of Energy and Resource Efficiency in Advanced Water Resource Recovery Facilities," U.S. Department of Energy, Research and Development for Advanced Water Resource Recovery Systems. DE-FOA-0002336. Awarded 2021. Total award \$2,400,000. co-Principal Investigator.

"Advancing Energy Resilience through Demonstrating, Characterizing, and Modeling Anaerobic Co-digestion of Organic Wastes for Widespread Implementation on DOD Installations." U.S. Department of Defense, Environmental Security Technology Certification Program. Total award \$925,000. co-Principal Investigator.

## PUBLICATIONS

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14. Grimm, T.R., **Newhart, K.B.** and Hering, A.S., “A Robust Self-starting Bayesian Approach for Multivariate Phase II Monitoring,” *Quality and Reliability Engineering International*, 2026, *in press*, DOI: 10.1002/qre.70213
13. **Newhart, K.B.**, Thompson, K.A., Branch, A., Saleson, A., Safarik, J., Plumlee, M.H., “NDMA soft-sensors for potable reuse: A model development study,” *Water Research*, 2025, 287, 124451, DOI: 10.1016/j.watres.2025.124451
12. Black, A., **Newhart, K.**, Linvill, C., Pytlar, A., Galaitsi, S., Fairfield, C., Wait, M., Bennett, E., Butkus, M., “A linguistic analysis of energy terminology in the wastewater literature,” *Water Science & Technology*, 2025, 91, 6, 671-683, DOI: 10.2166/wst.2025.036
11. Grimm, T.R., **Newhart, K.B.** and Hering, A.S., “Nonparametric Threshold Estimation of Autocorrelated Statistics in Multivariate Statistical Process Monitoring,” *Journal of Chemometrics*, 2025, 39, 2, DOI: 10.1002/cem.70004
10. Grimm, T.R., Branch, A., Thompson, K.A., Salevon, A., Zhao, J., Johnson, D., Hering, A.S., **Newhart, K.B.**, “Long-term Statistical Process Monitoring of an Ultrafiltration Water Treatment Process,” *ACS ES&T Engineering*, 2024, DOI: 10.1021/acsestengg.4c00042
9. **Newhart, K.B.**, Klanderman, M.C., Hering, A.S., Cath, T.Y., “A holistic evaluation of multivariate statistical process monitoring in a biological and membrane treatment system,” *ACS ES&T Water*, 2023, DOI: 10.1021/acsestwater.3c00058
8. Martin, M., Goethals, P., **Newhart, K.**, Rhodes, E., Vogel, J., Stevenson, B., “Optimization of sewage sampling for wastewater-based epidemiology through stochastic modeling,” *Journal of Engineering and Applied Science*, 2023, 70, 11.
7. **Newhart, K.B.**, Pfluger, A.R., Butkus, M.A., “The Green Escape Room: Part 2 – Teaching Students Professional Engineering Ethics by Applying Environmental Engineering Principles and Deciphering Clues and Puzzles.” Paper presented at *2022 ASCE Annual Conference & Exposition*, Minneapolis, MN, 2022.
6. **Newhart, K.B.**, Hering, A.S., Cath, T.Y., “Data science tools to enable decarbonized water and wastewater treatment systems.” *Pathways to Water Sector Decarbonization, Carbon Capture and Utilization*, edited by Z. Jason Ren and Krishna Pagilla, IWA Publishing, 2022.
5. **Newhart, K.B.**, Goldman-Torres, J., Wisdom, B., Freedman, D., Hering, A.S., Cath, T.Y., “Real-time dose control of peracetic acid disinfection in municipal wastewater treatment,” *ACS ES&T Water*, 2021, 1, 2, 328–338
4. **Newhart, K.B.**, Marks, C.A., Rauch-Williams, T., Cath, T.Y., Hering, A.S. “Hybrid statistical-machine learning ammonia forecasting in continuous activated sludge treatment for improved process control,” *Journal of Water Process Engineering*, 2020, 37, 101389
3. Klanderman, M., **Newhart, K.B.**, Cath, T.Y., Hering, A.S., “Fault isolation for a complex decentralized wastewater treatment facility,” *Journal of the Royal Statistical Society, Series C.*, 2020, 69, 931-951.
2. **Newhart, K.B.**, Holloway, R.W., Hering, A.S., Cath, T.Y., “Data-driven performance analyses of wastewater treatment plants: A review,” *Water Research*, 2019, 157, 498-513
1. Odom, G.J., **Newhart, K.B.**, Cath, T.Y., Hering, A.S., “Multi-state multivariate statistical process control,” *Applied Stochastic Models in Business and Industry*, 2018, 34(6), 880-892
0. Bell, E.A., Poynor, T.E., **Newhart, K.B.**, Regnery, J., Coday, B.D., Cath, T.Y., “Produced water treatment using forward osmosis membranes: evaluation of extended-time performance and fouling,” *Journal of Membrane Science*, 2017, 525, 77-88.

## WORKSHOPS

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“Planning Your Data and Analytics Roadmap” *WEFTEC*, September 28, 2025, Chicago, IL, Co-organizer

“An Introduction to Machine Learning Tools for Solving Environmental Challenges” *AEESP*, May 2025, Duke University, Organizer

“Doing More, with Less: Implementing Machine Learning Process Controls at WRRFs” *WEFTEC*, October 6, 2024, New Orleans, LA, Co-presenter

“Defining Pathways for Solving Environmental Challenges using Machine Learning” *AEESP Annual Conference*, May 2023, Organizer

“A Convergence of WRF Machine Learning Based Controller Implementation and Research” *WEF/IWA Innovations in Process Engineering*, June 6, 2023, Facilitator and presenter

“Visualization, Analysis, and Modeling in R for the Water Professional” *MoWaTER PRO: Data Science Workshop*, 2021 & 2022, Organizer

“Machine Learning in the Water Industry” *WEF/IWA Innovations in Process Engineering*, June 8, 2021, Organizer

“Understanding and Embracing Machine Learning, Artificial Intelligence and Predictive Analytics,” *AWWA/SWAN International Smart Water Symposium*, November 10, 2020, Facilitator and presenter

“Data Research Advances Water Industry,” *NSF Mid-scale Research Infrastructure Workshop for Intelligent Water Systems*, August 25, 2020, Virtual, Facilitator and presenter

## INVITED TALKS

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“AI & Digital Tools in Water Research” *National Alliance on Water Innovation Next Gen Seminar Series*, January 28, 2026, Webcast, Panelist

“Data-driven modeling in the water sector and the paradigm shift in predictive methods” *IWA Modeling and Integrated Assessment Seminar Series*, December 10, 2025, Webcast, Presenter

“Bridging Infrastructure and Intelligence: AI-Enhanced Water Resource Recovery,” *Marquette University*, November 12, 2025

“Emerging Challenges and Research Frontiers for Artificial Intelligence in Water and Wastewater Treatment,” *National Academy of Science: Water Science and Technology Board*, Frontier Applications of AI and Water Management, October 30, 2025, Presenter and Panelist

“From Data to Decisions: Why Machine Learning is Difficult but Essential for 21st Century Water Treatment Systems,” *Oregon State University*, October 24, 2025

“GenAI in Capstone,” *AEESP Education Committee*, Incorporating AI into the environmental engineering and science curriculum, Presenter, May 9, 2025, URL

“Digital Dreams: AI & the Water Sector,” New York’s Water Event, *NYAWWA*, April 15, 2025, Keynote Address

“Implementing Machine Learning Process Controls at Water Resource Recovery Facilities” *Water Research Foundation*, March 18, 2025, Webcast, Presenter

“Data-Driven Fault Protection and Process Control for Potable Reuse with Reverse Osmosis,” *National Alliance for Water Innovation*, January 22, 2025, Webcast, Presenter

“Digital Water: Leveraging Statistical and Machine Learning in Water Treatment Research,” *University of Michigan*, July 2024

“Machine Learning to the Rescue: Mining data to optimize water reclamation disinfection,” *Water Environment Federation*, June 13, 2024, Webcast, Presenter

“Process Modeling and Machine Learning – Selecting the Right Tool(s) for Troubleshooting and Optimization,” *Water Environment Federation*, November 30, 2023, Webcast, Presenter

“Artificial Intelligence in the Water Industry,” *Orange County Water District*, August 1, 2023, Webcast, Presenter, URL

“Understanding and Embracing Machine Learning, Artificial Intelligence and Predictive Analytics” *Metropolitan Water Reclamation District of Greater Chicago*, June 30, 2023, Seminar

“A Hypothetical – Potable Reuse Moves Towards Artificial Intelligence,” *36th Annual WaterReuse Symposium*, March 1, 2021, Panelist

## SELECT CONFERENCE PRESENTATIONS

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“From Machine Learning to Agentic AI: Exploring the Journey for Utilities and their Workforce” *WEFTEC*, 2025, Chicago, IL

“Prediction of Post-Secondary E. coli for Disinfection Control: Application of statistical and machine learning algorithms.” *WEF/IWA Innovations in Process Engineering*, June 9, 2023, Portland, OR

“Predictive Control in Wastewater Treatment Facilities Using Simple Statistical Models,” *South Platte Coalition for Urban River Evaluation: Confluence at the Confluence*, Oct 15, 2019, Englewood, CO

“Energy Reduction in Municipal Wastewater Treatment,” *Colorado Industrial Pretreatment Coordinators Association Fall Conference*, Oct 18, 2019, Black Hawk, CO

“Predictive Modeling and Performance Assessment of Ammonia-Based Aeration Control,” *WEFTEC*, Sept 23, 2019, Chicago, IL

“A Utility Perspective: Practical Considerations of Operating and Advancing Ammonia-Based Aeration Control,” July 10, 2019, *RMWEA Innovation Seminar*, Denver, CO

“Fault Detection Using PCA at a Municipal Wastewater Treatment Facility,” July 30, 2019, *Joint Statistical Meeting*, Denver, CO

“Performance Evaluation of a Sequencing Batch Membrane Bioreactor Using Principal Component Analysis,” *Annual Water Reuse Symposium*, Sept 11, 2017, Phoenix, AZ

“Use of Principal Component Analysis for Early-Fault Detection in a Pilot-Scale Biological Wastewater Treatment System,” *Quality and Productivity Research Conference*, June 14, 2017, Storrs, CT

## NON-REFEREED PUBLICATIONS

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Weintraut, Z., **Newhart, K.**, Thompson, K., Roostaei, J., “Are you ready for big data? A checklist for readiness for data analytics in water utilities,” *Journal AWWA*, 2022, 114, 10, 78-82

**Newhart, K.B.** & Avila, I., “NDMA: relevance and regulatory status for drinking water facilities,” *Rocky Mountain Water*, November 2017

## AWARDS

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ASCE Environmental Engineering Division Early Career Award, 2022

ACS Publications Peer Reviewer, Certificate of Recognition, 2022

WEF/WRF LIFT Intelligent Water System Challenge, 1st place, 2019

AWRA-Colorado Rich Herbert Memorial Scholarship, 2019

## LEADERSHIP

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Chair, National Water Research Institute (NWRI) Independent Expert Panel, supporting *Data-Driven Fault Detection and Process Control for Potable Reuse with Reverse Osmosis Project*

President, NSF ReNUWIt Engineering Research Center Student Leadership Committee, 2018 – 2019

President, CSM Campus Chapter of the Rocky Mountain Section of the American Water Works Association (RMSAWWA)/Rocky Mountain Water Environment Association (RMWEA), 2018 – 2019

Co-Chair, 15th Annual RMSAWWA/RMWEA Joint Student Conference, 2018

## SERVICE

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### Institution (OSU)

- Program Safety Committee, September 2024– present

### Professional

- Early Career Board, ACS Environmental Science & Technology Engineering, January 2024 – present
- Editorial Advisory Team, ACS Environmental Science & Technology Water, June 2022 – present

- Referee, ACS Environmental Science & Technology Engineering; Environmental Science: Water Research & Technology; Resources, Conservation & Recycling; Water Environment Research; Water Research (IWA); Journal of Water & Health; Springer Nature: Scientific Reports
- Technology Reviewer, Water Research Foundation TechLink, 2022
- Member, AWWA Water Science & Research Division, Information Management & Technology, 2021 – 2023

## IN THE NEWS

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Newhart, K. B., Marks, C., Rauch-Williams, T., Cath, T. Y., Hering, A. S. (2020) “Boulder tests its waters with predictive aeration control,” *Advances in Water Research*, 30: 25–28. URL.

## CERTIFICATIONS

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Wastewater Operator, Class D, Colorado, 2016-2024

Fundamentals of Engineering (FE), Environmental, Colorado, NCEES ID 16-475-7

## TEACHING

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| <i>Institution</i> | <i>Course</i>            | <i>Title (Credit Hours)</i>                               | <i>Terms</i> |
|--------------------|--------------------------|-----------------------------------------------------------|--------------|
| OSU                | ENVE 490 <sup>1</sup>    | Environmental Engineering Design (3)                      | 1            |
| OSU                | ENVE 415 <sup>1</sup>    | Environmental Engineering Laboratory (3)                  | 1            |
| USMA               | EV201 <sup>1</sup>       | Introduction to Environmental Engineering (3)             | 2            |
| USMA               | EV350 <sup>2</sup>       | Environmental Engineering Technologies (3)                | 1            |
| USMA               | EV401 <sup>1</sup>       | Physical and Chemical Treatment (3.5)                     | 3            |
| USMA               | EV450 <sup>2</sup>       | Environmental Engineering for Sustainable Development (3) | 3            |
| USMA               | EV490/491 <sup>2</sup>   | Environmental Engineering Design (Capstone) (3)           | 5            |
| USMA               | XE365 <sup>3</sup>       | Advanced Experimental Methods & Data Processing (3)       | 1            |
| RRCC               | WQM42 <sup>1</sup>       | Water Data Management & Analysis (3)                      | 1            |
| CSM                | CEE 470/570 <sup>3</sup> | Unit Processes for Water and Wastewater Treatment (3)     | 3            |
| CSM                | CEE 471/571 <sup>3</sup> | Advanced Water Treatment and Reclamation (3)              | 1            |
| CSM                | CEE 330 <sup>3</sup>     | Field Session for Environmental Engineering (3)           | 3            |

<sup>1</sup> Course director, <sup>2</sup> Instructor, <sup>3</sup> TA / Guest lecturer

## ADVISEES

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### Primary Advisor

Perry Wilson, MS, Environmental Engineering, 2024-Present

### Committee Member

Matthew Latham, PhD, Environmental Engineering, 2023-Present

Sampson Achagwe Antwi, PhD, Environmental Engineering, 2024-Present

Sofia Zook, PhD, Environmental Engineering, Colorado School of Mines, 2024-Present

Taylor Grimm, PhD, Statistical Science, Baylor University, 2021-2024